Reading between the lines: LLM-based label imputation and meta-learning for noisy labels in emergency department triage

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Abstract. Children presenting to the SickKids pediatric emergency department wait twice: once for physician assessment, and again for diagnostic test results. The Machine Learning Medical Directive (MLMD) system eliminates the second wait for high-confidence cases by predicting required tests from triage-time features (Canadian Triage and Acuity Scale (CTAS) score, vital signs, chief complaint, and demographics) and authorizing them in parallel. Each missed prediction returns the patient to the slower, serialized workflow, making recall the primary performance target. Label noise constrains this recall. Tests ordered at external facilities go unrecorded in the SickKids electronic health record (EHR), causing genuine test-ordering events to appear as negatives and teaching the model spurious clinical absences.

We address this across 483,705 pediatric ED encounters spanning June 2018 to January 2026. At the data level, we replace the production regex imputation system with a Knowledge Graph and GraphRAG retrieval pipeline, validated through a multi-agent clinical audit, that identifies 11,110 candidate false negatives across four diagnostic pathways. At the training level, we apply LiLAW (Lightweight Learnable Adaptive Weighting) [1], a meta-learning framework that down-weights residual false negatives through bilevel optimization over three learned scalar parameters.

GraphRAG identified 11,110 candidate false negatives, achieving precision of 0.944 on the most semantically ambiguous pathway (Testicular Ultrasounds) against 0.048 for lexical matching. On tabular benchmarks with asymmetric noise matching the MLMD structure, LiLAW improved PR-AUC by 1.2–2.9 percentage points across 10–40% noise rates, with gains concentrated at the high-precision operating point clinically relevant to triage deployment. The downstream ablation integrating both interventions with the full MLMD architecture is in progress; component-level validation establishes the expected performance envelope and confirms the noise correction mechanisms function as designed.

Keywords: *label noise, clinical decision support, machine learning, label imputation, meta-learning*

Introduction and Clinical Motivation

Redundant wait times constrain patient throughput in the pediatric Emergency Department (ED) at The Hospital for Sick Children (SickKids). Under standard care, patients queue twice: first for an initial physician assessment to order diagnostics, then again for results. Faster diagnostic turnaround enables earlier treatment decisions, shortens time to disposition, and reduces ED crowding—factors that directly affect patient outcomes and experience.

The hospital has implemented a Machine Learning Medical Directive (MLMD) system that uses triage data to predict and authorize relevant tests proactively, parallelizing diagnostic processing with the initial wait [2]. The MLMD targets four high-volume, time-sensitive diagnostic pathways: arm X-rays, electrocardiograms (ECGs), appendix ultrasounds, and testicular ultrasounds. However, the model's performance across these pathways is limited by significant label noise. Positive labels in the SickKids Electronic Health Record (EHR) are reliable, but negative labels are ambiguous because the EHR is not integrated with external providers. Because records of externally performed tests are unavailable, their occurrence must be inferred retrospectively from the patient's final documented diagnosis. The current baseline system utilizes a regex-driven approach to identify variations of diagnoses in clinical text and map them to their corresponding required tests. While this rules-based mapping is clinician-validated, it is inherently brittle; it struggles to capture the vast semantic nuance, shorthand, and variability inherent to clinical documentation. As a result, many valid diagnoses—and the external tests they imply—are missed by the regex, causing the system to misclassify these instances as true negatives. Across 483,705 triage encounters spanning June 2018 to January 2026, this label ambiguity pushes the MLMD toward a conservative decision boundary that artificially depresses recall. Each relevant test the model fails to authorize returns a patient to the slower, serialized workflow, forfeiting substantial efficiency gains.

To overcome this limitation, we propose a two-pronged methodology to handle asymmetric label noise at both the data and algorithmic levels. First, we replace the brittle regex baseline with a dynamic, evidence-grounded Knowledge Graph and Large Language Model (LLM) pipeline, capable of semantically parsing clinical triage notes to accurately impute missing test labels—recovering 11,110 test-ordering events the regex baseline missed. Second, to manage the residual uncertainty inherent to imputed labels, we apply LiLAW (Lightweight Learnable Adaptive Weighting) [1], a meta-learning framework that dynamically down-weights suspected false negatives during training. By combining semantic knowledge extraction with robust meta-learning, we demonstrate that the MLMD can safely expand its decision boundary, substantially improving recall and unlocking further efficiency gains for the pediatric ED.

Related Work

Label imputation in EHRs. Missing labels in clinical datasets are rarely missing at random; their absence reflects institutional silos [3] and clinical workflow patterns [4]. Most imputation work targets missing continuous *features* (vital signs, laboratory values) [5], but imputing missing *labels* is a distinct problem. The SickKids noise structure fits the Positive-Unlabeled (PU) learning paradigm [6]: positive labels are trustworthy, while the unlabeled class conflates true negatives with unobserved external events. Prior PU approaches treat the unlabeled class as a statistical quantity to estimate, whether through Bayesian mixture models [7] or label-recovery frameworks for clinical codes [8]. Our approach resolves the ambiguity directly: we query a validated Knowledge Graph against each patient's post-assessment diagnosis before training begins, converting a probabilistic inference problem into a clinical evidence lookup where the graph permits it.

Knowledge graphs and LLMs in clinical decision support. LLMs show strong capacity for medical knowledge imputation [9] and clinical entity recognition from unstructured notes [10], but standalone LLM outputs lack auditability and hallucinate in high-stakes settings. Retrieval-augmented generation (RAG) over clinical guidelines improves extraction accuracy by grounding outputs in validated sources [11, 12]. GraphRAG architectures [13, 14] extend this by combining structured Knowledge Graph reasoning with contextual retrieval, enabling semantically coherent inference across interconnected clinical concepts. We adopt a GraphRAG architecture but depart from existing work in three ways: the Knowledge Graph is constructed dynamically from live clinical guidelines, enriched with ontology-standardized codes and quantitative literature evidence, and every diagnosis-to-test pathway is validated through a multi-agent clinical audit before use.

Meta-learning for noisy labels. Early approaches to learning with noisy labels modified the loss function itself. Loss-correction methods estimate a class-conditional noise transition matrix and use it to adjust either the loss or the model's predictions, yielding an unbiased estimator of the clean risk under ideal conditions [15]. Generalized Cross Entropy [16] and Symmetric Cross Entropy [17] avoid explicit matrix estimation by interpolating between noise-tolerant and optimization-friendly loss families. These methods improve robustness when corruption follows class-level flip rates, but they impose a fixed rule at the loss level and cannot adapt per-sample trust as training progresses.

Sample selection methods addressed this limitation by choosing which examples to learn from at each step. MentorNet [18] learns a data-driven curriculum through an auxiliary network, typically requiring clean supervision to train the mentor. Co-teaching [19] removes the clean-data assumption by training two peer networks that exchange small-loss examples, at roughly twice the backbone cost. DivideMix [20] extends this idea by fitting a two-component Gaussian mixture to per-sample losses and recasting suspected-noisy examples as unlabeled data for semi-supervised co-training. These methods exploit the empirical finding that deep networks fit clean patterns before memorizing noise, but low loss is an imperfect proxy for label correctness. Under asymmetric corruption, genuinely difficult examples can resemble mislabeled ones, and the heuristic breaks down.

Meta-learning reweighting made the selection rule itself learnable through bilevel optimization. Learning to Reweight Examples (L2RW) [21] optimizes one scalar weight per training example in each mini-batch to minimize loss on a small clean validation set, at roughly three times standard training cost. Meta-Weight-Net [22] replaces per-example weights with a small multilayer perceptron that maps loss to weight, but retains the clean-validation-set requirement. Confident Learning [23] takes a complementary approach, estimating the joint distribution of noisy and true labels to prune errors before training rather than reweight during it. LiLAW [1] compresses the meta-learner to three global scalar parameters (α, β, δ) that weight easy, moderate, and hard samples based on softmax agreement and prediction confidence. It retains bilevel updates against validation mini-batches but does not require those labels to be clean or unbiased, and adds no auxiliary networks. These properties match the SickKids setting, where a small trustworthy validation set may not exist and the noise structure (unrecorded external tests producing false negatives, not symmetric random flips) differs from the corruption patterns most benchmark methods were designed for.

Data and Setting

The dataset comprises EHR records from pediatric patients (aged 0–18 years) presenting to the SickKids ED between June 2018 and January 2026. After excluding patients who left without being seen (n = 25,397) and encounters missing the free-text clinical diagnoses required for label imputation (n = 11,396), the final cohort contained N = 483,705 unique ED encounters (median age 5.0 years, IQR 2.0–10.0; 54.1% male).

The model's feature space is limited to information available at triage: demographics (age, sex, weight), vital signs (pulse, respiration, temperature, blood pressure), chief complaint narrative, and Canadian Triage and Acuity Scale (CTAS) score. Post-assessment ICD-10 (International Classification of Diseases, 10th Revision) diagnosis codes, assigned by the treating physician after evaluation, are used only by the label imputation pipeline and excluded from the model's input features to prevent target leakage.

Four binary test-ordered labels define the prediction targets: Arm X-Rays (n = 1,904 positive encounters), ECGs (n = 12,236), Appendix Ultrasounds (n = 3,293), and Testicular Ultrasounds (n = 2,889). Positive labels are reliable ground truth: they record tests ordered and performed at SickKids. Negative labels are ambiguous. A recorded absence may reflect a genuine clinical decision not to test, or a test performed at an external facility whose records never reach the SickKids EHR. This asymmetric noise structure is the central problem the methods below address.

Methods[1]

Label imputation via Knowledge Graph

The label imputation pipeline constructs an evidence-grounded Knowledge Graph (KG) mapping ICD-10 diagnosis codes to clinically indicated diagnostic tests. When a patient's post-assessment diagnosis maps to a test but the encounter lacks a positive label for that test, the label is imputed positive. Imputation runs offline over the training set only; no diagnosis codes are available at inference time.

Knowledge Graph construction. The graph is built in four stages. GPT-5-mini generates initial triage rules grounded via RAG over the EPFL clinical guidelines dataset. All entities are then standardized to recognized medical ontologies: symptom nodes receive HP and MONDO terms; condition nodes receive WHO ICD-10 and ICD-10-CA codes; test nodes receive LOINC panel codes. Edges are weighted with literature co-occurrence counts (Europe PMC) and clinical trial counts (ClinicalTrials.gov) to distinguish well-supported associations from marginal ones. Full API credentials, endpoint details, and resulting schema columns are documented in Appendix B. Finally, ClinGraph [24], a 153K-node, ~3M-edge clinical knowledge graph spanning SNOMED CT, ICD-10-CM, LOINC, and RxNorm, back-fills missing cross-vocabulary codes and adds up to 500 first-degree neighbor nodes per existing node.

Multi-agent audit. Before imputation, every diagnosis-to-test pathway is validated by a four-persona audit pipeline. Three independent personas evaluate each triad: a Protocol & Triage Enforcer with RAG access over clinical guidelines, a Rapid Diagnostics Researcher querying live literature databases, and an ED Attending evaluating clinical flow pragmatics. A fourth persona, the ED Medical Director, synthesizes these assessments into a verdict (*Verified*, *Flagged for Review*, or *Rejected*) and a confidence score in $[0, 1]$. Pathways with genuine clinical errors, Choosing Wisely flags, or zero evidence support are unconditionally flagged. Audit outcomes are written back as edge attributes. Full persona specifications, weighting logic, hard-flag triggers, and checkpointing details are provided in Appendix C.

Imputation strategies. We evaluate two retrieval strategies. *KG String Matching* uses bidirectional substring matching to map patient diagnosis strings to KG nodes, then traverses 1-hop and 2-hop edges to identify indicated tests. A minimum-node threshold τ requires at least τ independently matched nodes before imputation, reducing false positives from broad symptom matches (Algorithm 1). *GraphRAG Semantic Retrieval* embeds KG nodes and patient diagnoses using the MedEmbed biomedical language model against a ChromaDB vector store to capture cases without lexical overlap. A sliding-window negation filter discards matches where the clinical concept is negated (e.g., "no chest pain" does not trigger ECG imputation), and indicated tests are ranked by aggregate cosine similarity and filtered by a threshold θ (Algorithm 2).

Noise-aware training via LiLAW

Label imputation corrects false negatives where a post-assessment diagnosis conclusively implies a test was needed. It cannot resolve ambiguous cases: chest pain sometimes warrants an ECG and sometimes does not. Residual false negatives remain in both the original and imputed training sets. LiLAW (Lightweight Learnable Adaptive Weighting) [1] handles this residual noise by learning to down-weight samples whose observed label is inconsistent with the model's predictions.

Sample difficulty and weight functions. Let $s_i = \text{softmax}(f_\theta(x_i))$ be the model's output for training sample i with observed label $\tilde{y}_i$. LiLAW characterizes each sample by the predicted probability of the observed label $s_i[\tilde{y}_i]$ and peak class confidence $\max(s_i)$, defining three regimes:

- *Easy*: $s_i[\tilde{y}_i] = \max(s_i)$ and $\max(s_i)$ is high. The model agrees with the observed label confidently.
- *Moderate*: $\max(s_i)$ is low regardless of agreement. The model is uncertain.
- *Hard*: $s_i[\tilde{y}_i] < \max(s_i)$ and $\max(s_i)$ is high. The model confidently predicts a different class — the signature of a likely mislabelled sample.

[1] Access to the production codebase was granted late in the project timeline. Some experimental conditions described here were not completed in time; results are reported only where validation was fully executed.

Hard samples correspond precisely to the false negatives that remain after imputation: encounters where the model assigns high probability to "test needed" but the training label records "test not performed." LiLAW introduces three scalar meta-parameters, α, β, and δ, each governing a weight function:

$$\mathcal{W}_\alpha = \sigma(\alpha \cdot s_i[\tilde{y}_i] - \max(s_i)) \tag{1}$$

$$\mathcal{W}_\beta = \sigma(-(\beta \cdot s_i[\tilde{y}_i] - \max(s_i))) \tag{2}$$

$$\mathcal{W}_\delta = \exp\left(-\frac{(\delta \cdot s_i[\tilde{y}_i] - \max(s_i))^2}{2}\right) \tag{3}$$

$\mathcal{W}_\alpha$ is a sigmoid high for easy samples; $\mathcal{W}_\beta$ is an inverted sigmoid high for hard samples; $\mathcal{W}_\delta$ is a radial basis function centered on the moderate regime. The total weight $\mathcal{W} = \mathcal{W}_\alpha + \mathcal{W}_\beta + \mathcal{W}_\delta$ scales the loss: $\mathcal{L}_W = \mathcal{W} \cdot \mathcal{L}$. Learning α, β, and δ allows the model to adapt which difficulty regime to emphasize at each training stage.

Bilevel optimization. Meta-parameters are learned via bilevel optimization. Each mini-batch alternates two steps: model parameters θ update on the weighted training loss $\mathcal{L}_W$ with meta-parameters frozen; then θ is frozen and the meta-parameters update by SGD on a validation mini-batch. The joint objective is:

$$\begin{aligned} \theta^* &= \arg\min_\theta \sum_{(x_i,\tilde{y}_i)\in\mathcal{D}_t} \mathcal{L}_{W_{\alpha^*,\beta^*,\delta^*}} \\ \text{s.t.} \quad \alpha^*,\beta^*,\delta^* &= \arg\min_{\alpha,\beta,\delta} \sum_{(x_j,\tilde{y}_j)\in\mathcal{D}_v} \mathcal{L}_{W_{\theta^*}} \end{aligned} \tag{4}$$

The validation set $\mathcal{D}_v$ does not need to be clean. Accuracy on a held-out noisy set is an unbiased estimator of accuracy under the noisy distribution [1], so LiLAW steers training using the same validation partition as early stopping, a critical property where a clean-labelled EHR subset is unavailable. Meta-parameters initialize at $\alpha = 10.0$, $\beta = 2.0$, $\delta = 6.0$ and update by SGD (lr = 0.005, weight decay 10^{-4}) after a single warmup epoch of standard BCE loss.

Combined approach

In the combined condition, LiLAW training is applied to the GraphRAG-imputed dataset. The two interventions target complementary failure modes. GraphRAG corrects the deterministic subset of false negatives, where a documented diagnosis clearly implies a test was needed. LiLAW handles the residual probabilistic uncertainty, down-weighting samples whose loss magnitude during training suggests misclassification. Imputation reduces the noise rate; LiLAW adapts to whatever noise remains.

Evaluation[1]

Data partitioning

The cohort is partitioned temporally to simulate prospective deployment: encounters are sorted by arrival date and split sequentially so the model is always evaluated on data from later in time than its training window. Random splitting would allow future information to leak into training through seasonal and longitudinal patterns in the data. Following the protocol of the predecessor study [2], 55% of encounters form the training set, 15% the validation set, and 30% the held-out test set, spanning June 2018 to January 2026.

Model architecture

All four experimental conditions train and evaluate the same *MultiModalFusionTransformer* architecture. Tabular features (vitals, demographics, CTAS score, postal distance, cyclical datetime encodings) project from a 27-dimensional input to a 256-dimensional embedding. Three free-text fields are each encoded by ClinicalBERT [25]

to 256-dimensional embeddings. The four representations pass through four transformer encoder layers (4 heads, feedforward dimension 1024), an attention pooling layer, and a linear classification head, totalling approximately 3.3M parameters. This replaces the 5-layer fully connected network from the original MLMD validation [2] and uses free-text triage notes directly without concept extraction.

Training configuration

All conditions use identical hyperparameters to isolate the effect of label strategy: RAdam (lr 10^{-3}, weight decay 10^{-4}) with ReduceLROnPlateau scheduling, up to 15 epochs with early stopping on validation loss, batch size 64. Class imbalance is handled via a per-pathway positive weight $w^+ = \min\left(\frac{N_{\text{neg}}}{N_{\text{pos}}},\ 10.0\right)$. Positive-class proportions range from 0.4% (Arm X-Rays) to 2.5% (ECGs), producing raw imbalance ratios between 38:1 and 252:1 before capping.

Experimental conditions

We compare four training conditions:

1. *Production Baseline*: standard training on original regex-imputed labels.
2. *GraphRAG Imputation Only*: standard training on the KG-imputed dataset (11,110 additional positive labels).
3. *LiLAW Only*: LiLAW-weighted training on original regex labels.
4. *Combined*: LiLAW-weighted training on the KG-imputed dataset.

Both pipeline components have been implemented and validated in isolation (Section 6); the integrated four-condition ablation on the full cohort is the immediate next evaluation step.

Metrics

Recall is the primary metric: each missed prediction returns a patient to the serialized queue. Precision is the safety constraint: unnecessary authorizations burden ancillary departments and erode clinician trust. We report full precision-recall (PR) curves rather than single-threshold summaries, since clinical deployment requires understanding the efficiency-safety tradeoff before selecting an operating point. AUROC captures discriminative ability under class imbalance. Expected Calibration Error (ECE) measures whether confidence scores are reliable enough for threshold selection based on stated probability rather than empirical tuning.

Imputation evaluation

Novel positive imputations absent from the production regex baseline are classified as "discoveries": candidate false negatives the regex system missed. A clinical domain expert performed retrospective chart review on a random sample of newly imputed labels to validate that discoveries reflect genuine false negatives; results are reported in Section 6.

Results

Label imputation performance

Table 1 compares the two imputation strategies against the regex baseline across all four diagnostic pathways.

KG String Matching showed extreme variance. It performed well on Appendix Ultrasounds (F_1 = 0.801) where documentation is standardized, but failed on Arm X-Rays (Recall: 0.031) and Testicular Ultrasounds (Precision: 0.048, 20,035 discoveries) where anatomical terms overlap with unrelated symptom descriptions.

GraphRAG Semantic Retrieval was more consistent: 0.944 precision on Testicular Ultrasounds and 0.733 recall on Arm X-Rays, recovering presentations that lexical matching missed. The exception was Appendix Ultrasounds (recall 0.198), where the embedding space flattens distinctions that substring matching preserves in this narrowly documented pathway. GraphRAG identified 11,110 total discoveries.

Table 1. Performance of both imputation strategies against the regex baseline. "Discoveries" are novel positive imputations absent from the regex baseline; in this Positive-Unlabeled setting they represent potential clinical false negatives.

Pathway	Method	Precision	Recall	F_1	Discoveries
ECG	KG String	0.786	0.871	0.827	2,900
ECG	GraphRAG	**0.817**	0.837	**0.827**	2,292
Arm X-Ray	KG String	0.009	0.031	0.013	6,696
Arm X-Ray	GraphRAG	**0.139**	**0.733**	**0.234**	8,632
Appendix US	KG String	0.809	**0.793**	**0.801**	617
Appendix US	GraphRAG	**0.870**	0.198	0.322	97
Testicular US	KG String	0.048	0.352	0.085	20,035
Testicular US	GraphRAG	**0.944**	**0.521**	**0.672**	89

LiLAW proxy validation

Tabular asymmetric noise benchmarks. We validated LiLAW on three tabular binary classification benchmarks under synthetically injected asymmetric positive-to-negative noise, mirroring the MLMD structure. A 5-layer MLP was trained at four noise rates (10–40%) with noise confined to train and validation splits; results are averaged over three seeds with a 55/15/30 split.

Adult Income (48,842 samples, 14 features) most closely matches the MLMD problem: mixed continuous and categorical features with moderate class imbalance. LiLAW improved PR-AUC by 1.2–2.9 percentage points across all four noise rates, with gains generally scaling with noise rate (Table 2).

Table 2. LiLAW proxy validation on Adult Income with asymmetric positive-to-negative noise. PR-AUC and Recall at PPV ≥ 0.80 (Rec@PPV80), mean ± std over 3 seeds.

Noise Rate	Baseline PR-AUC	LiLAW PR-AUC	Δ PR-AUC	Baseline Rec@PPV80	LiLAW Rec@PPV80
10%	0.692 ±0.019	0.702 ±0.017	+1.3%	0.276 ±0.090	0.296 ±0.101
20%	0.691 ±0.016	0.699 ±0.018	+1.2%	0.295 ±0.074	0.294 ±0.093
30%	0.686 ±0.014	0.700 ±0.019	+2.1%	0.295 ±0.052	0.297 ±0.088
40%	0.679 ±0.015	0.699 ±0.018	+2.9%	0.290 ±0.060	0.311 ±0.081

At the Rec@PPV80 operating point, LiLAW recovered 2.1 additional recall percentage points at 40% noise (0.311 vs. 0.290 baseline). On Breast Cancer, directional improvements of 2–3% appeared, but one seed consistently collapsed the model, inflating variance and masking the signal. On Pima Indians (768 samples), gains were marginal at low noise and turned negative at 30–40%; the meta-update batches are too small at this dataset size to reliably steer the meta-parameters.

These gains fall well short of the LiLAW paper's headline results (+17.2% Top-1 accuracy at 50% symmetric noise on CIFAR-100-M [1]). Two structural differences explain why: asymmetric one-directional noise conflates hard-noisy with hard-clean samples, weakening the difficulty signal LiLAW exploits, and the Adult baseline degrades only modestly across noise rates (0.692 to 0.679 PR-AUC), limiting headroom. The 1–3% range is the expected operating envelope for LiLAW in the MLMD setting.

MedMNIST replication. We replicated the BloodMNIST experiment (11,959 training samples, 8 classes, symmetric noise) with ResNet-18, ImageNet pretraining, and early stopping (patience=10) over 30 epochs. LiLAW did not consistently outperform the baseline: accuracy differences were within ±0.5% at all noise rates (Table 3). The 0% baseline reached 96.4%, confirming the setup is sound.

Meta-parameter diagnostics identified the cause. α saturated near its maximum ($\approx$10) within the first few epochs regardless of noise rate, exhausting the easy-sample suppression signal early. β and δ did adapt monotonically with noise (increasing by 0.6 and 1.1 from 0% to 80%), confirming the weighting mechanism responds to corruption. The most natural explanation is epoch budget: patience=10 terminated training before the β and δ dynamics could reshape the loss landscape. A follow-up sweep extends training to 100 epochs on BloodMNIST and DermaMNIST at 0%, 30%, and 50% noise; results are pending.

Table 3. BloodMNIST replication under symmetric noise, ResNet-18 with ImageNet pretraining. Mean Top-1 accuracy over 3 seeds.

Noise Rate	Baseline Accuracy	LiLAW Accuracy	Δ
0%	96.39%	96.74%	+0.35%
20%	94.47%	94.13%	−0.35%
40%	92.94%	92.78%	−0.16%
60%	90.15%	89.62%	−0.53%
80%	82.09%	82.59%	+0.50%

Downstream MLMD integration

Both pipeline components have been implemented against the full MLMD architecture. The four-condition ablation across the 483,705-encounter cohort encountered configuration issues from late system access and is not complete at submission. The proxy validation establishes the expected operating range: 1.2–2.9% PR-AUC improvement under tabular asymmetric noise, with gains scaling with noise rate and concentrated at the high-precision operating point. Whether that magnitude transfers to the multi-modal transformer, and whether imputation and LiLAW provide complementary or overlapping recall gains, are the questions the pending ablation will answer.

String matching requires negligible overhead (0.55 ms per encounter, 6.16 ms initialization). GraphRAG incurs higher cost (175.54 ms per encounter, 17.75 s initialization) but this latency is inconsequential for offline imputation. A phase-level breakdown separating entity matching from graph traversal is provided in Appendix E.

Discussion and Limitations

The GraphRAG pipeline identified 11,110 candidate false negatives the production regex system missed. The divergence between retrieval strategies is instructive: GraphRAG achieved 0.944 precision on Testicular Ultrasounds where string matching collapsed to 0.048, because biomedical embeddings resolve the anatomical term overlap that overwhelms substring matching. The inverse held for Appendix Ultrasounds, where documentation is sufficiently standardized that lexical matching outperforms semantic generalization (0.793 vs. 0.198 recall). A pathway-conditional strategy would combine the strengths of both.

The proxy validation establishes that LiLAW improves ranking quality by 1.2–2.9% PR-AUC under the target noise conditions, with gains scaling with noise rate. The pending ablation will determine whether imputation and LiLAW provide complementary recall gains on the full MLMD cohort. Expected magnitude is in the 1–3% range demonstrated in proxy validation, not the 17% reported on CIFAR; the MLMD setting offers less headroom because the baseline degrades only modestly and noise is likely below 40%.

The MLMD is decision support, not automation. Clinicians retain final authority to override any prediction. A missed prediction does not mean a missed test; the physician can still order it during assessment. The cost of a miss is time (the patient returns to the serialized workflow), not patient safety. Precision failures carry a different risk: unnecessary authorizations burden ancillary departments and erode clinician confidence over time. The evaluation framework reflects this asymmetry, treating recall as the efficiency metric and precision as the safety constraint.

Validation accuracy does not guarantee adoption. Triage staff need to see which clinical associations drove each prediction and challenge those that conflict with their judgment. *Diagnotix* provides this transparency as a clinical-facing web application where clinicians can explore the full Knowledge Graph interactively, generate new diagnostic pathways from live clinical guidelines, and interrogate any pathway through an evidence-cited chat interface. At 175 ms per encounter, the GraphRAG pipeline adds no perceptible latency in interactive use. Server initialization takes 17.7 s once at spin-up. Full architecture and API specifications are in Appendix F.

Limitations

The Knowledge Graph is a point-in-time artifact with no automated mechanism for incorporating guideline updates; a graph built on 2025 evidence may diverge from practice without continuous maintenance. Sensitivity to alternative biomedical encoders was not characterized. GraphRAG cannot recover cases where patients were

transferred before any diagnosis was assigned or where the SickKids assessment differed from the external facility's. LiLAW's effectiveness on real clinical asymmetric noise, as opposed to synthetic benchmarks, is evaluated here for the first time without external institutional validation. Neural network opacity imposes a practical ceiling on clinician adoption: the model cannot explain its predictions in clinical terms, limiting the depth of oversight it can support.

Future Work

The most immediate priority is making the Knowledge Graph self-maintaining: a continuous integration pipeline that periodically re-scrapes guideline sources and re-indexes the vector database, paired with a lightweight approval interface where new rule batches require clinician sign-off before entering the live graph. This would close the human-in-the-loop gap without manual reconstruction.

With maintenance in place, the natural next step is transitioning from offline label correction to live prospective decision support. The KG pipeline could infer needed diagnostics from presenting symptoms before physician assessment, directly serving the MLMD's parallelization objective. The GraphRAG pipeline's 175 ms per-encounter latency is acceptable at SickKids volumes but would require quantization or distilled encoders to scale across multi-facility deployments. At that scale, a 10-point recall improvement across SickKids' 80,000 annual visits redirects roughly 8,000 authorizations per year to the parallelized pathway. Asymmetric label noise from external clinical events is a systematic problem across fragmented hospital networks; the Knowledge Graph and meta-learning framework developed here offers a deployable solution that does not require cross-institutional EHR integration.

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Appendix A. Contributions

A breakdown of team contributions, encompassing both implementation and manuscript authorship, is provided in Table 4. At the request of collaborators at SickKids, the project's source code is hosted privately at `https://github.com/biswurmt/mlmd-noise/`. We plan to grant the instruction team access upon receipt of their GitHub handles.

Table 4. Detailed breakdown of teammate contributions to this report and underlying investigation.

Section	Contributors
Introduction	Tyler: narrative structure and clinical framing Alice: technical enrichment and editing
Related work	Tyler: literature review and source curation across both tracks; synthesis and final write-up for noise-resistant training track Alice: literature review, source discovery, synthesis, and final write-up for label imputation track
Data & setting	Alice: cohort selection, exclusions, and feature space description
Methods	
Label imputation	Alice: knowledge graph construction, multi-agent audit pipeline, and imputation strategies Tyler: ideation and design (RAG strategy over EPFL dataset, bidirectional KG prediction, etc.)
Noise-resistant training	Tyler: LiLAW implementation; proxy experimental design and validation; ablation analysis
Diagnotix	Alice: knowledge graph visualization, Navigate Mode, and dynamic pathway generation Tyler: Analyze Mode chatbot, grounded via RAG over embedded clinical chunks
Results	
KG validation & discoveries	Alice: KG clinical validation and discovery of clinical false negatives
MLMD performance	Alice: impact of label imputation Tyler: proxy validation and combined efficacy
Discussion	Alice: primary synthesis and label imputation interpretation Tyler: noise-resistant training discussion and limitations
Future work	Alice

Appendix B. Ontology Enrichment via External APIs

To ensure interoperability and standardize clinical entities, the Knowledge Graph build pipeline queries eight external services to attach standard medical codes and evidence metadata to each extracted record. Table 5 outlines this sequential enrichment process; step numbers correspond to the nine-step build pipeline described in the Methods section.

Table 5. API integration steps, targeted ontologies, and resulting dataset schema modifications.

Step	API (Target Ontology)	Credential	Enriched Entity	Column(s) Added
2	EMBL-EBI OLS4 (HP / MONDO / EFO)	None	`raw_symptom`	`ebi_name`, `ebi_code`, `synonyms`
3	Canada Health Infoway FHIR (SNOMED CT CA)	OAuth2 client credentials	`raw_symptom`	`infoway_name`, `infoway_code`
4	UMLS REST API (LOINC)	`UMLS_API_KEY`	`test` (unique values only)	`loinc_code`
5	UMLS REST API (ICD-10, WHO international)	`UMLS_API_KEY`	`condition` (unique values only)	`icd10_code`
6	Canada Health Infoway FHIR (ICD-10-CA)	OAuth2 client credentials	`condition` (unique values only)	`icd10ca_code`
7	NLM RxNorm & OpenFDA	None	`treatment` (where present)	`rxcui`, `adverse_event`
8	Europe PMC REST API	None	`symptom` → `condition` (`INDICATES_CONDITION`)	`literature_articles`, `literature_patents`
9	Europe PMC REST API	None	`condition` → `test` (`REQUIRES_TEST`)	`literature_articles`, `literature_patents`
10	Europe PMC REST API	None	`symptom` → `test` (`DIRECTLY_INDICATES_TEST`)	`literature_articles`, `literature_patents`
11	ClinicalTrials.gov v2	None	`condition` → `treatment` (`RECOMMENDS_TREATMENT`)	`trial_count`

Appendix C. Multi-Agent Graph Audit Pipeline

Audit Unit and Triad Extraction

The unit of audit is a clinical triad: a (finding → condition → test) path extracted by traversing `INDICATES_CONDITION` followed by `REQUIRES_TEST` edges. Three-hop paths are preferred over `DIRECTLY_INDICATES_TEST` shortcuts because they carry the condition context necessary for clinical validation. Triads are deduplicated on (finding, test) pairs and sorted by clinical acuity (HIGH → MODERATE → STANDARD) to prioritise the most time-critical pathways first.

Persona Descriptions

Four personas operate sequentially on each triad. Personas A–C produce independent assessments with no shared context at inference time; Persona D synthesizes their outputs into a final verdict.

Persona A — Protocol & Triage Enforcer. The only persona with external tool access. Persona A queries a local ChromaDB vector database built from the EPFL clinical guidelines dataset, retrieving the top-5 most relevant passages via semantic search filtered by test type. Outputs include a structured `support_level` (*strong* | *moderate* | *weak* | *none*), a matched guideline reference, whether a validated clinical decision rule is required before ordering, and a Choosing Wisely flag.

Persona B — Rapid Diagnostics Researcher. Operates without an LLM call. Persona B queries Semantic Scholar for literature co-evidence on the condition–test pair (with Europe PMC as fallback), returning paper count, mean publication year (classified as *current* ≥ 2020, *aging* ≥ 2015, *outdated* < 2015), and keyword flags indicating whether point-of-care alternatives or negative predictive value evidence appear in retrieved abstracts.

Persona C — ED Attending. A pure LLM call with no tool access. Persona C evaluates clinical flow pragmatics: over-testing risk, under-testing risk, safe discharge logic, wrong-test detection, and whether the pathway constitutes never-event mitigation — i.e., whether the test is mandated to catch a catastrophic, time-sensitive diagnosis.

Persona D — ED Medical Director. Receives the full structured outputs of Personas A, B, and C and synthesizes a final `audit_status` (*Verified* | *Flagged for Review* | *Rejected*) and a `confidence_score` $\in [0, 1]$. The weighting scheme is explicit: Persona A carries the dominant weight; Persona B adjusts confidence by up to ± 0.15 based on literature recency and paper count; Persona C's `pathway_is_never_event_mitigation` flag adds $+0.10$, while `wrong_test_concern` is a hard-flag trigger regardless of other scores.

Hard-Flag Triggers

Three conditions unconditionally produce a *Flagged for Review* verdict, bypassing the scoring logic:

1. Persona C raises `wrong_test_concern`, indicating a genuine clinical error in the Knowledge Graph.
2. Persona A raises a Choosing Wisely flag.
3. A rule-out pathway has zero guideline support and zero literature evidence simultaneously.

Audit Metadata and Checkpointing

Audit results are written back to the graph in-place. Each audited edge receives `audit_status`, `confidence_score`, `last_audited`, persona-level sub-scores, and `audit_rationale` as edge attributes, making results queryable during GraphRAG retrieval. A complete `audit_report.json` is also produced containing per-triad detail and a summary of verified, flagged, and rejected counts with mean confidence. To manage LLM call volume over a fully enumerated graph, completed triad results are cached and skipped on re-invocation via a `-resume` flag.

Appendix D. Detailed Imputation Methodologies

Knowledge Graph Traversal Algorithm

Algorithm 1 formalizes the deterministic substring matching and graph traversal procedure. The bidirectional substring check captures clinical shorthand and abbreviations; 1-hop and 2-hop traversal identifies both directly indicated tests and condition-mediated test requirements.

Algorithm 1 Knowledge Graph Entity Matching and Test Imputation

Require: Patient diagnosis string D, Knowledge Graph $G = (V, E)$, minimum node threshold τ
Ensure: Set of imputed diagnostic tests P

```
T ← Split(D, ";")                                   ▷ Isolate terms from compound diagnosis
M ← ∅                                               ▷ Matched KG nodes
Step 1: Bidirectional Substring Matching
for each t ∈ T do
    t ← Lowercase(Strip(t))
    for each v ∈ V where Type(v) ∈ {Condition, Symptom, Risk_Factor, ...} do
        L ← Lowercase(Label(v))
        if t ⊆ L or L ⊆ t then
            M ← M ∪ {v}
        end if
    end for
end for
Step 2: Graph Traversal and Test Voting
TestCounts ← empty map                              ▷ Unique nodes indicating each test
for each m ∈ M do
    TestsForM ← ∅
    for each outgoing edge (m, u) ∈ E do
        if Relation(m, u) ∈ {REQUIRES_TEST, DIRECTLY_INDICATES_TEST} then
            TestsForM ← TestsForM ∪ {u}                                  ▷ 1-hop
        else if Relation(m, u) = INDICATES_CONDITION then
            for each outgoing edge (u, w) ∈ E do
                if Relation(u, w) = REQUIRES_TEST then
                    TestsForM ← TestsForM ∪ {w}                          ▷ 2-hop
                end if
            end for
        end if
    end for
    for each test ∈ TestsForM do
        TestCounts[test] ← TestCounts[test] + 1
    end for
end for
Step 3: Threshold Filtering
P ← {test | TestCounts[test] ≥ τ}
return P
```

Semantic Retrieval and Score Aggregation

Algorithm 2 formalizes the GraphRAG semantic pipeline. Patient diagnoses are embedded using the MedEmbed biomedical language model and queried against a ChromaDB vector store. Before accepting a match, a six-token negation window scans the original text for tokens such as *no, not, denies, without, absent, negative for, rules out*; matches under negation are discarded. Instead of discrete voting, each matched node contributes its cosine similarity score to the tests it indicates, and tests are filtered by an aggregate score threshold θ.

Algorithm 2 GraphRAG Semantic Retrieval and Score-Based Imputation

Require: Patient diagnosis D, Knowledge Graph $G = (V, E)$, Vector Store $\mathcal{V}$, thresholds $k, s_{\min}, \theta$
Ensure: Set of imputed diagnostic tests P

```
T ← Split(D, ";")                                   ▷ Isolate terms from compound diagnosis
M ← empty map                                       ▷ Max cosine score per matched KG node
Step 1: Vector Search and Negation Filtering
for each t ∈ T do
    E_t ← Embed(t)
    Candidates ← QueryVectorStore(V, E_t, top_k = k)
    for each c ∈ Candidates do
        score ← 1.0 − c.distance                    ▷ Distance to cosine similarity
        if score ≥ s_min and not IsNegated(D, c.label, window = 6) then
            M[c.node] ← max(M.get(c.node, 0), score)
        end if
    end for
end for
Step 2: Graph Traversal and Score Aggregation
TestScores ← empty map
for each (m, score) ∈ M do
    TestsForM ← ∅
    for each outgoing edge (m, u) ∈ E do
        if Relation(m, u) ∈ {REQUIRES_TEST, DIRECTLY_INDICATES_TEST} then
            TestsForM ← TestsForM ∪ {u}
        else if Relation(m, u) = INDICATES_CONDITION then
            for each outgoing edge (u, w) ∈ E do
                if Relation(u, w) = REQUIRES_TEST then
                    TestsForM ← TestsForM ∪ {w}
                end if
            end for
        end if
    end for
    for each test ∈ TestsForM do
        TestScores[test] ← TestScores.get(test, 0) + score
    end for
end for
Step 3: Aggregate Thresholding
P ← {test | TestScores[test] ≥ θ}
return P
```

Appendix E. Computational Latency Benchmarks

To isolate the computational bottleneck within the GraphRAG semantic pipeline, benchmarking separated inference into two phases: entity matching (embedding generation and vector search) and graph traversal (score aggregation and test imputation). Table 6 reports mean latency across 50 iterations for the matching phase and 25 iterations for the full end-to-end pipeline.

The overwhelming majority of latency in the semantic pipeline occurs during Phase 1, driven by the computational cost of the local biomedical embedding model. The subsequent graph traversal (Phase 2) adds only ∼37 ms overhead, confirming that the bottleneck is the encoder rather than graph structure.

Table 6. Computational latency benchmarks comparing the baseline lexical string matching strategy against the GraphRAG semantic retrieval pipeline.

Metric	String Matching	Semantic (GraphRAG)
Cold-Start Initialization		
KG Pickle / Model Load	6.16 ms	17,745.32 ms
Per-Encounter Latency		
Phase 1: Entity Matching Only	0.42 ms	137.92 ms
Phase 2: Full Pipeline (w/ Traversal)	0.55 ms	175.54 ms

Appendix F. Diagnotix Web Application Architecture

Diagnotix is built on a React 19, TypeScript, and Vite frontend coupled with a Python 3.12 FastAPI backend. The system supports LLM integration through both Nebius (Kimi-K2.5-fast) and Azure OpenAI. The application is divided into two primary operational modes: *Navigate* and *Analyze*.

Navigate Mode

Navigate Mode is the default visual exploration interface for the clinical Knowledge Graph.

- **Interactive Graph Canvas.** A force-directed 2D visualization powered by `react-force-graph-2d`.
- **Pathway Picker.** Filters the visualization to the 1-hop subgraph of a selected diagnostic test (e.g., ECG, Appendix Ultrasound). An "All Pathways" option reveals the full macroscopic graph structure.
- **Node Search & Legend.** Facilitates node search by label with an interactive legend for toggling node-type visibility.
- **Dynamic Pathway Generation.** Users input a novel diagnostic test (e.g., CT Head), triggering a backend pipeline that grounds itself via RAG over Qdrant and live Semantic Scholar abstracts to generate 8–15 triage rules (up to 3 convergence iterations). New rules are appended to `guideline_rules.json` and the Knowledge Graph is rebuilt; only the delta (new nodes and edges) is returned to the canvas, avoiding full page reloads.

Analyze Mode

Analyze Mode provides a conversational interface for a selected diagnostic pathway.

- **Contextual Injection.** The selected pathway's subgraph (nodes and edges) is injected directly into the LLM context window.
- **Evidence-Grounded Generation.** Responses from Nebius Kimi-K2.5-fast are enriched by RAG retrieval from clinical guidelines (Qdrant) and live Semantic Scholar abstracts; all outputs are cited.
- **State Management.** Multi-turn conversation history is preserved independently per pathway.

Backend API Endpoints

Table 7. Diagnotix FastAPI backend endpoints.

Endpoint	Purpose
`GET /api/tests`	Retrieve a list of all diagnostic test nodes.
`GET /api/graph?pathway=<name>`	Retrieve the full graph or a specific 1-hop pathway subgraph.
`POST /api/add_test`	Execute the LLM pipeline to generate rules; return the node/edge diff.
`POST /api/chat`	Initialize clinical chat with RAG and Semantic Scholar grounding.